



Eitekh WorkOS

Enterprise Work Operating System & Methodology Governance

VERIFIED SYSTEM SPECIFICATION DOCUMENT

Technical & Functional Platform Specifications

A complete architectural blueprint, security matrix, module breakdown, and presentation guide for enterprise stakeholders and clients.

DOCUMENT ID	PLATFORM EDITION	VERIFICATION HEALTH	ENGINE RUNTIME
EIT-SPEC-2026-V2	Enterprise v2.0.0	Clean (596/596 Tests Passed)	Next.js 16 (Turbopack)

**Verified Quality Certification:** This document reflects the verified, production-compiled codebase of Eitekh WorkOS. Static type analysis is 100% verified (0 errors on Next.js 16/Turbopack), 40 Jest test suites passed, 148 API endpoints validated with Zod, and real-time Server-Sent Events (SSE) active.

1. Executive Summary & Value Proposition

Eitekh WorkOS is an all-in-one work management and issue tracking operating system engineered for enterprise agility, compliance, and large-scale digital transformations. It combines modern agile execution frameworks (Scrum, Kanban, Gantt Roadmaps) with native **SAP Activate Methodology Governance**, strict **Policy-Based Access Control (PBAC)**, and real-time Server-Sent Events (SSE) data synchronization.

Strategic Enterprise Differentiators

- **Zero-Latency Synchronization:** Event-driven Server-Sent Events (SSE) keep all distributed team members synchronized across boards, backlogs, and charts without manual page reloads.
- **Native SAP Activate Engine:** Industry-first native implementation of the standard 6-phase SAP Activate methodology (Fit-to-Standard workshops, WRICEF backlog generator, and 4-eyes Quality Gate sign-offs).
- **Rigorous Security Governance:** 6-tier Policy-Based Access Control (PBAC), AES-256-GCM field encryption, and 100% input schema validation across all 148 API endpoints.
- **Boardroom-Ready Reporting:** 10 deep-dive analytical views with instant one-click export into formatted multi-page PDF, Microsoft Excel (.xlsx), and standard CSV formats.

2. Technical Architecture & System Specifications

The platform adheres to modern cloud-native architectural patterns, separating presentation, business logic, access control, and persistence layers into decoupled, observable modules.

Architecture Layer	Technology Component	Enterprise Specifications & Standards
User Interface & Presentation	Next.js 16 (Turbopack) + React 19	Server/Client component splitting, WCAG-compliant accessibility, native zero-flicker Dark/Light themes, responsive viewports.
Interactive Framework	@dnd-kit (Core & Sortable)	Hardware-accelerated drag-and-drop for Kanban boards, sprint backlogs, and priority ranking with full keyboard accessible navigation.
Backend Application Layer	Next.js API Engine (Node.js)	148 standardized RESTful route handlers with consistent JSON error handling, execution telemetry, and structured audit logs.
Input Validation & Sanitization	Zod 4.6 (100% Coverage)	Centralized schema validation on all inputs, query bounds, payload caps, and XSS URL sanitization (blocks javascript: schemes).
ORM & Persistence	Prisma ORM 5.22	Strongly typed relational schema with 24 relational foreign-key indexes, transactional integrity (\$transaction), and tenant isolation.
Security & Authorization	Hierarchical PBAC Engine	6-tier hierarchy: Viewer (10) → Member (20) → Team Lead (30) → Project Admin (40) → Org Admin (50) → Super Admin (60).
Cryptography & Encryption	AES-256-GCM & bcrypt	Field-level encryption for webhook secrets and tokens; 10-salt bcrypt password hashing; SHA-256 OTP hashing with atomic attempt limits.
Real-Time Messaging Bus	Server-Sent Events (SSE)	Event-driven reactive sync engine with automatic client reconnection, heartbeat ping, and configurable sync intervals.
Defense-in-Depth Headers	HTTP Security Middleware	Strict Content Security Policy (CSP), HSTS (63072000s), X-Frame-Options: DENY, X-Content-Type-Options: nosniff, CSRF origin verification.

3. Security, Authentication & PBAC Governance

Eitekh WorkOS enforces a multi-tier defense architecture ensuring zero data leakage between organizations and complete accountability.

Role Level	Role Key	Access Scope	Permitted Operations
Level 10	VIEWER	Assigned Projects	Read-only inspection of permitted boards, tasks, roadmaps, and reports. Cannot modify issues.
Level 20	MEMBER	Assigned Projects	Standard developer/contributor: create issues, transition assigned tasks, add attachments, post comments.
Level 30	TEAM_LEAD	Team & Workspace	Manage team backlogs, prioritize sprints, assign issues, run sprint ceremonies, balance team workloads.
Level 40	PROJECT_ADMIN	Project Scope	Configure custom project workflows, status transitions, custom fields, components, versions, and project memberships.
Level 50	ORG_ADMIN	Organization Scope	Workspace provisioning, organization-wide user invitations, billing management, organization audit logs.
Level 60	SUPER_ADMIN	Platform Scope	Multi-tenant oversight, cross-organization audits, security threat operations, system cache management, and platform health telemetry.

4. Functional Capabilities & Interactive Views

The platform provides 8 specialized core workspace views, allowing users to adapt their operational style according to their role and project methodology:

Workspace View	Primary Purpose	Key Interactive Capabilities
 Kanban Board	Visual Workflow Execution	Drag-and-drop cards across columns, Work-In-Progress (WIP) limit thresholds with visual overload warnings, quick search and priority filters.
 Interactive List	High-Density Task Triage	Dense data grid with multi-selection checkboxes and a floating Bulk Actions Bar for batch status changes, priority shifts, and batch assignment.
 Scrum & Backlog	Iterative Sprint Planning	Drag items between backlog and active sprint, story point velocity tallies, sprint lifecycle controls (Plan → Start → Complete Sprint).
 Gantt / Timeline	Strategic Roadmapping	Dynamic date ranges, priority color bars, interactive milestone dependencies with SVG connector lines, and today marker indicators.
 Calendar View	Date-Driven Milestone Tracking	Month-by-month grid, quick "Today" jump, due-date highlights, multi-issue day count badges, and click-to-preview drawers.
 Team Workload	Resource Capacity Balancing	Calculates team member utilization bandwidth, flags over-allocated team members (>100% capacity), and assists in load rebalancing.
 SAP Activate	ERP Implementation Engine	6-phase methodology rail (Discover → Run), Fit-to-Standard workshop evaluator, automated WRICEF backlog generator, and 4-eyes Q-Gates.
 Charts & Analytics	Operational Intelligence	Live Velocity metrics, Cumulative Flow Diagrams (CFD), Defect Leakage, Lead & Cycle time distributions, and Live Sync indicator deck.

5. Native SAP Activate Implementation Methodology

A premier enterprise differentiator built natively into Eitekh WorkOS for SAP S/4HANA, SuccessFactors, and ERP digital transformations:



Module Component	Methodology Purpose	Platform Functionality
Fit-to-Standard Workshops	Explore Phase Solution Design	Interactive evaluation of standard SAP scope items with 6 decisions: ADOPT, CONFIGURE, EXTEND, INTEGRATE, DEFER, OUT_OF_SCOPE.
Automated Backlog Generator	Sprint Readiness & Traceability	Decisions to "Extend" or "Integrate" automatically generate tracked WRICEF development tickets and configuration tasks directly in the sprint backlog.
Quality Gates (Q-Gates)	Phase-Exit Formal Governance	Multi-criteria sign-off gate between phases enforcing strict separation of duties (4-eyes principle) between Gate Raiser and Gate Approver.
Pre-Packaged Scope Catalogues	Accelerated Blueprinting	Seeded accelerators including SuccessFactors and S/4HANA standard scope items, deliverables, and role RACI templates.

6. Executive Analytics & Multi-Format Reporting Center

The platform features a dedicated Reports Center containing 10 enterprise reports with on-demand instant export capabilities:

Report Title	Target Audience	Analytical Insights Delivered
1. Sprint Velocity & Forecast	Product Managers & Scrum Masters	Commitment vs completed story points, rolling velocity average, and future milestone delivery projections.
2. Burndown & Burnup Tracking	Engineering Leads	Real-time scope changes, ideal vs actual trajectory, and mid-sprint risk early detection.
3. Cumulative Flow Diagram (CFD)	Agile Coaches & PMO	Work-In-Progress stability, process bottlenecks, queue buildup, and lead time trends across all workflow columns.
4. Lead & Cycle Time Distribution	VP of Engineering & Directors	Average time from issue conception to delivery (Lead Time) and active coding to completion (Cycle Time).
5. Bug & Defect Leakage Analysis	Quality Assurance (QA) Directors	Defect escape rate by component, severity distribution, root cause categories, and mean time to resolution (MTTR).
6. Epic Progress & Completion	Portfolio Directors & Executives	High-level strategic initiative completion percentages, remaining child task estimation, and target date variance.
7. Team Workload & Allocation	Resource & Department Managers	Individual contributor hours logged, story points assigned, and bandwidth capacity threshold utilization.
8. Issue Stagnation & Aging	Project Managers	Identifies forgotten tasks, blockers sitting in status without activity for >7 days, and triage compliance.
9. Traceability & Dependency Matrix	Compliance & Enterprise Auditors	Full upstream and downstream linkage mapping: Epic → User Story → Sub-Task → Bug with blocker detection.
10. Project SLA Compliance Audit	Client Relationship & Delivery Leads	First-response time, priority-based resolution SLA adherence, and contract breach risk monitoring.

7. Client Demonstration Walkthrough Script

Follow this curated 6-stage presentation narrative to showcase technical sophistication and business ROI in under 10 minutes:

Stage 1: The Executive Command Center (1 min)

Log in and highlight the clean interface. Trigger the global **Command Palette** (Ctrl + K) to show instant fuzzy search across tasks, projects, and users. Toggle between Dark and Light themes to highlight design polish.

Stage 2: Agile Fluidity & Real-Time Sync (2 min)

Open an active project. Drag an issue across the Kanban board columns. Highlight the green pulsing **Live Synced** badge in the header, explaining that all team members see updates in real time with zero manual page refreshes.

Stage 3: High-Density Triage & Bulk Actions (1 min)

Switch to List View. Select three tasks with checkboxes. Demonstrate the floating **Bulk Actions Bar** updating priorities or statuses in one batch action.

Stage 4: Roadmaps & Workload Capacity (1.5 min)

Switch to the Gantt Timeline to show quarterly milestones and visual dependency lines. Open the Workload tab to demonstrate team bandwidth tracking and over-allocation warning thresholds.

Stage 5: The Enterprise Showstopper — SAP Activate (2.5 min)

Click into the **SAP Activate** tab. Walk through the 6 phase gates. Conduct a live Fit-to-Standard workshop assessment (Adopt vs Extend), show the auto-generated WRICEF tickets in the backlog, and explain the Quality Gate dual-approval governance.

Stage 6: The Executive Closer — Boardroom Exports (1 min)

Navigate to the Reports Center. Open the *Sprint Velocity Report*, click **Export to PDF**, and show the stakeholder-ready formatted document, ready for immediate C-level presentation.